MUHAMMAD IMRAN	BT503 Environment Biotechnology
Past papers solved questions	MID TERM

I. What is the bio indicator?

Bioindicators are living organisms such as plants, planktons, animals, and microbes, which are utilized to screen the health of the natural ecosystem in the environment. The expression 'Bioindicator' is used as an aggregate term referring to all sources of biotic and abiotic reactions to ecological changes.They are used for assessing environmental health and biogeographic changes taking place in the environment e.g., a plankton responding rapidly to changes taking place in the surrounding

2. Importance of an aerobic zone?

It is essential to have anaerobic and aerobic zones in the treatment line for the biological removal of phosphorus. The anaerobic zone gives good conditions for the development or selection of a large population of phosphorus accumulating organisms.

Bio filter? What are the bio filter give example?

Biofiltration is a pollution control technique using a bioreactor containing living material to capture and biologically degrade pollutants.

Examples of biofiltration include: Bioswales, biostrips, biobags, bioscrubbers, Vermifilters and trickling **filters**. Constructed wetlands and natural wetlands. Slow sand **filters**.

3. Glyphosate?

'Glyphosate' (herbicide) is an analogue of phosphoenol pyruvate and inhibits the enzyme 5-enolpyruvylshikimate-3-phosphate synthase. The gene coding for this enzyme has been identified, isolated and inserted into a number of plants
e.g., Petunias. The gene was expressed behind a CaMV promoter and introduced using A. tumefaciens.

4. How we improve crop quality?

Consider these 9 tips, techniques, and methods regarding how to increase your corn crop yield

1. Plant Early, Plant Effectively. ...
2. Practice Seasonal Soil Rotation. ...
3. Know The Yield Potential. ...
4. Always Scout Your Fields. ...
5. Ensure Proper Water Drainage. ...
6. Utilize Fertilizers. ...
7. Test Your Soil. ...
8. Weed Early and Often.

5. Two advantage of conventional system of sludge?

Advantages of Conventional Dispersed Plug Flow Process 1. Allows smaller volume than CSTR 2. Flexible operation, zone aeration, step feeding options, accommodates anoxic and aerobic processes with single biomass for biological nutrient removal 3. Less aeration than CSTR 4.

6. How marine plant indicate the health of environment give example?

Marine plants provide valuable information to predict the status of oceanic environment, as they are immobile and rapidly obtain equilibrium with their natural surrounding.
Changes in the diversity of species of phytoplankton, including Euglena clastica, Phacus tortus, and Trachelon anas, indicate the pollution of marine ecosystems

7. What are phase of bio filters?

The first methods to be developed. Consists of a relatively large vessel or container, typically made of cast concrete, metal or durable plastic, which holds a filter medium of organic material such as peat, heather, bark chips and the like.The gas to be treated is forced, or drawn, through the filter.

8. Three phase of bio filter?

This biofilter is a three-phase reactor composed of: Solid phase: consists of a support medium and biofilms, Liquid phase: consists of the liquid in permanent flow through the porous medium, Gas phase: formed by artificial aeration and by the gaseous by-products of the biological activity.

9. Importance of anaerobic zone?

The anaerobic zone gives good conditions for the development or selection of a large population of phosphorus accumulating organisms.

10. Transformation of plant resistance against herbicide?

The transfer of **herbicide resistance** genes from various sources into crop **plants** has been performed using several techniques. These transgenic products rely on both target site and metabolic detoxification **resistance** mechanisms.

11. Acidophile?

Acidophiles or **acidophilic** organisms are those that thrive under highly acidic conditions (usually at pH 2.0 or below). These organisms can be found in different branches of the tree of life, including Archaea, Bacteria, and Eukarya.

12. Phytoplankton?

Phytoplankton are microscopic marine algae. **Phytoplankton** is the base of several aquatic food webs. In a balanced ecosystem, they provide food for a wide range of sea creatures including whales, shrimp, snails, and jellyfish.

13. Function of bio trickling filter over bio filter?

A **biotrickling filter** is a combination of a **biofilter** and a bioscrubber. The bacteria responsible for decomposition are immobilised **on** a carrier or **filter** material. ..... This water also carries away excess sludge/biofilm, as well as decomposition products which may hinder the biomass.

14. Role of phytoplankton for the transfer of pollutant among the different trophic level? What is the role of phytoplankton in the transfer of pollutants among different trophic level?

Phytoplankton are the primary producers **in** aquatic ecosystems and play a key **role in** the **transport** of organic contaminants through the **food chain** to higher **trophic levels .....** The bioaccumulation at lower **trophic levels** forms the first step **in** the **transfer** of chemicals through the **food chain**.

15. Advantage of bio indicator?

The most important reasons for using **bioindicators** are: the direct determination of **biological** effects, the determination of synergetic and antagonistic effects of multiple pollutants on an organism, the early recognition of pollutant damage to plants as well as toxic dangers to humans and.

16. Submerged bio filter?

Besides being a support medium for biomass growth, the granular material acts also as a filter medium. Periodic washings are necessary to eliminate the excess biomass accumulated.

17. Biotechnology and its all applications?

Biotechnology has **applications** in four major industrial areas, including health care (medical), crop production and agriculture, non-food (industrial) **uses** of crops and other products (e.g. biodegradable plastics, vegetable oil, biofuels, and environmental **uses**).

18. Pollutant?

A pollutant is a substance or energy introduced into the environment that has undesired effects, or adversely affects the usefulness of a resource.

19. Bio reactor?

a bioreactor is a vessel in which a chemical process is carried out which involves organisms or biochemically active substances derived from such organisms.

20. Glysophates?

Glyphosate is a broad-spectrum systemic herbicide and crop desiccant. It is an organophosphorus compound, specifically a phosphonate, which acts by inhibiting the plant enzyme 5-enolpyruvylshikimate- 3-phosphate synthase. It is used to kill weeds, especially annual broadleaf weeds and grasses that compete with crops.

21. List two advantage of conventional activated sludge system?

High biomass concentration in the reactor due to the recirculation of the settled bacteria from the bottom of the secondary sedimentation tank. Biomass remains in the system longer than the liquid, resulting in high BOD removal efficiency.

22. Environmental biotechnology?

Environmental biotechnology is biotechnology that is applied to and used to study the natural environment. Environmental biotechnology could also imply that one try to harness biological process for commercial uses and exploitation.

23. How can be prepared plant as recombinant?

In simpler terms, **recombinant DNA** is formed when the **DNA** from one organism is incorporated into the **DNA** of another organism of the same or different species ..... Using enzymes like ligase, the cut **DNA** fragment is attached to the vector. This is then introduced into the host cell, which takes it up as part of its own **DNA**.

23. How plants are modified as bio indicator?

Plant indicators The presence or absence of certain **plant** or other vegetative life in an ecosystem can provide important clues about the health of the environment: environmental preservation. There are several types of **plant** biomonitors, including mosses, lichens, tree bark, bark pockets, tree rings, and leaves.

24. What are thermophile? Where are they found?

A thermophile is an organism—a type of extremophile—that thrives at relatively high temperatures, between 41 and
122 °C. Many thermopiles are archaea. Thermophilic eubacteria are suggested to have been among the earliest bacteria.**Thermopiles** are **found** in various exothermally heated regions of the Earth, such as hot springs like those in Yellowstone National Park (see image) and deep sea hydrothermal vents, as well as decaying plant matter, such as peat bogs and compost.

25. Animal as bio indicator and give example?

A **bioindicator** is a living organism that **gives** us an idea of the health of an ecosystem. ... The numbers of earthworms in the soil can also be used to indicate the health of the soil. **One example** of a **bioindicator** is lichens.

26. Genetic modification of popular plants how they remove mercury from plants? 5

Genetic modification of **plants** involves adding a specific stretch of DNA into the plant's genome, giving it new or different characteristics. This could include changing the way the plant grows, or making it resistant to a particular disease.

There are two types of **mercury removal** materials: non-regenerative sorbents and regenerative adsorbents. With non-regenerative types of **mercury** sorbents, the process fluid flows through the sorbent bed for a number of years, after which the sorbent is replaced.

27. Difference between transformation, transduction and conjugation? 5

In **transformation**, a bacterium takes up a piece of DNA floating in its environment. In **transduction**, DNA **is** accidentally moved from one bacterium to another by a virus. In **conjugation**, DNA **is** transferred **between** bacteria through a tube **between** cells.

28. Function of filter medium in bio filter and of compound used in? 5

The medium offers good water holding capacity and soluble chemicals within the waste gas dissolve into the film of moisture around the matrix. The medium itself provides physical support for microbial growth, with a large surface area to volume ratio, high in internal void spaces and rich in nutrients to stimulate and sustain bacterial activity

29. Recombinant microorganism particularly bacteria and yeast? 5

GE of microorganisms for use in environmental biotechnology: expansion of metabolic pathways either to modify the existent metabolic capability, or to introduce new pathways.

Many examples… A strain of Escherichia coli into which was engineered some 15 genes originating from Pseudomonas.

These were introduced to construct a pathway able to produce indigo for the dyeing of denim (Bialy, 1997).

The traditional method requires the use of toxic chemicals.

30. What Are The Two Main Responsibilities Of DNA Vector?2

Viruses may be used as vectors i.e., the DNA of viruses can be used to carry foreign DNA. Most such recombinant viral DNA vectors need phage particles to perform proper gene transfer i.e. transduction. Transfer of naked viral recombinant vector to host cells is called transfection, and tends to have lower uptake rates as compared to transduction.

30. What Is Xenobiotics? 2

XENOBIOTICS‘ are the foreign chemical substances found within an organism, which are not normally naturally produced by an organism or expected to be present within that organism.

31. What Are Vectors? 2

a **vector** is an organism that does not cause disease itself but which spreads infection by conveying pathogens from one host to another. ... A "**vector**" in this sense is a vehicle for delivering genetic material such as DNA to a cell

32. Define Ion Exchange Chromatography? 2

on chromatography is a chromatography process that separates ions and polar molecules based on their affinity to the ion exchanger. It works on almost any kind of charged molecule—including large proteins, small nucleotides, and amino acids.

33. What Is Biosensor? 2

A biosensor is an analytical device, used for the detection of a chemical substance, that combines a biological component with a physicochemical detector.

34. What Is Recombinant? 2

Recombinant DNA molecules are DNA molecules formed by laboratory methods of genetic recombination to bring together genetic material from multiple sources, creating sequences that would not otherwise be found in the genome. **(Internet)**

35. Environmental Biotechnology? 2

Environmental biotechnology is the branch of biotechnology that addresses environmental problems, such as the removal of pollution, renewable energy generation or biomass production, by exploiting biological processes (nature.com)

36. Bioaccumulation? 2

—The collection of substances e.g. toxins and chemicals, in an organism‘s body is called bioaccumulation‖. Sources: substances from any source e.g. air, water, soil, could be stocked through bioaccumulation.

37. Use Of Recombinant Viruses?

A **recombinant virus** is a **virus** produced by recombining pieces of DNA using **recombinant** DNA technology. This may be used to produce **viral** vaccines or gene therapy vectors. It is also used to refer to naturally occurring **recombination** between **virus** genomes in a cell infected by more than one **virus** strain. **(Internet)**

38. What Is Mobility Of DNA? 2

Movement of genes within and between organisms. Genetic transfer through physical contact between bacteria

(conjugation). Genetic transfer through bacteriophages (Transduction). Taking up of foreign DNA by bacteria (Transformation).

39.What Is Thermophiles And Their Application In Genetic Engineering? 2

Thermophiles are organisms that grow and thrive at temperatures (60– 80°C) that are often too high for mesophiles—most thermophiles are Archaea.

Possible Application of Advancement of Thermophilic Bacterial Research Wastes utilization Biomass conversion

Biofuels Food bioprocessing Detergents.

40.Define Puc8 Plasmid? 3

pUC8 Plasmid The **pUC8 plasmid** was designed by scientists and contains the lac z gene. To produce the **plasmid**, the pBR322 **plasmid** was cut in half with EcoR I and the section containing the ampicillin resistance gene was combined with a DNA fragment containing the lac z gene. **(Internet)**

41.What Is Toxicity? 3

Toxicity is the degree to which a chemical substance or a particular mixture of substances can damage an organism. Toxicity can refer to the effect on a whole organism, such as an animal, bacterium, or plant, as well as the effect on a substructure of the organism, such as a cell (cytotoxicity) or an organ such as the liver (hepatotoxicity).

42.What Is PCR? 3

Polymerase chain reaction (**PCR**) is a method widely used in molecular biology to make several copies of a

specific DNA segment. Using **PCR**, copies of DNA sequences are exponentially amplified to generate thousands to millions of more copies of that particular DNA segment.

43.Endocrine Disruptor Examples? 3

E.g., synthetic oestrogens (such as 17αethinyloestradiol forming active ingredient of birth control pills) and natural oestrogens.

E.g., feminisation of male fish in many species including minnows, trout and flounders.

44.Cosmid Vector? 3

A **cosmid** is a type of hybrid plasmid that contains a Lambda phage cos sequence . **Cosmids** (cos sites + plasmid = **cosmids**) DNA sequences are originally from the lambda phage. They are often used as a cloning **vector** in genetic engineering. **Cosmids** can be used to build genomic libraries.

45.Define Biotechnology? 3

Biotechnology: “The application of science and technology to living organisms, as well as parts, products and models thereof, to alter living or non-living materials for the production of knowledge, goods and services” (OECD, 2002). There was a time when the biotechnology sector was seen as largely medical or pharmaceutical in nature, particularly amongst the general public… The agricultural, industrial and environmental applications of biotechnology are potentially enormous.

46.Enzyme Equipment And Solution Used In Genetic Engineering? 3

Here is a list of a genetic engineer's molecular tools/enzymes most commonly used in genetic engineering experiments:

1. Polymerase Chain Reaction (PCR) ...
2. Restriction **Enzymes** (Molecular Scissor) ...
3. Gel Electrophoresis. ...
4. DNA Ligase. ...
5. Plasmids. ...
6. Transformation/Transduction. ...
7. Identifying Transgenic Organisms.

47.PLac Z Expression Vector? 3

Lac –Z promoter operator is in frame with lac-Z alpha fragment (the NH3 terminal part of Galactosidase gene.

Multiple cloning sites are found in the border of NH3 end including ATG sequence.

48.Why Does Hydrogen Bonding Occur? 3

he reason **hydrogen bonding occurs** is because the electron is not shared evenly between a **hydrogen** atom and a negatively charged atom. **Hydrogen** in a **bond** still only has one electron, while it takes two electrons for a stable electron pair. ... Any compound with polar covalent **bonds** has the potential to form **hydrogen bonds**.

49.Catabolic Pathways? 3

This is a **catabolic pathway**. **Catabolic pathways** involve the degradation (or breakdown) of complex molecules into simpler ones. Molecular energy stored in the bonds of complex molecules is released in **catabolic pathways** and harvested in such a way that it can be used to produce ATP.

50.Principle Of Plant Transformation? 3

transformation is the geneticalteration of a cell resulting from the direct uptake and incorporation of exogenous genetic material from its surroundings through the cell membrane.

Principle: There are number of principles which are used for the transformation of the plants which are given: Vector method: There are two vector methods which are given below: Agrobacterium Viral transformation Vector Free method: There are different vector free methods which are given below: Gene gun Electroporationt.

51.Steps Of cDNA Libraries? 3

cDNA libraries In eukaryotes, the first product of transcription from DNA is not messenger RNA (mRNA) but heterogeneous nuclear RNA (hnRNA). This is mRNA prior to the removal of all the non-coding sections, or introns, which are discarded during the processing to produce the mature mRNA.

cDNA is DNA which has been artificially made using the mature mRNA as a template (through Reverse Transcriptase enzyme).

52.Detail Note On Recombinant Microorganisms Bacteria And Yeast? 5

Recombinant bacteria is a type of bacteria which have the genetic material DNA of the another organism. Recombinant DNA is a technology which is developed by scientists that made it possible to insert a gene into the genetic material of a bacterium. This “recombinant” micro-organism can now produce such type of protein that coded by the gene. The scientists made the human insulin gene in the laboratory by using this technology.

The yeast in which introduce the characteristics according to our desire is called the recombinant yeast. Yeasts are unicellular eukaryotic microbial host cells that offer unique advantages in producing biopharmaceutical proteins.

53.Applications of Cloning Vectors? 5

Method of gene **cloning** is useful in studying the structure and function of genes in detail. Medical **Applications**: In medicine, **cloned** bacteria plays important role for the synthesis of vitamins, hormones and antibiotics. Agricultural **Applications**: **cloning** in Bacteria facilitates nitrogen fixation in plants.

54.Note On Zooplankton? 5

Zooplankton are heterotrophic plankton. Plankton are organisms drifting in oceans, seas, and bodies of fresh water.

The word "zooplankton" is derived from the Greek zoon, meaning "animal", and planktos, meaning "wanderer" or

"drifter".

Zooplankton is a categorization spanning a range of organism sizes including small protozoans and large metazoans. It includes holoplanktonic organisms whose complete life cycle lies within the plankton, as well as meroplanktonic organisms that spend part of their lives in the plankton before graduating to either the nekton or a sessile, benthic existence zooplankton are primarily transported by different mechanism which are given below: ambient water currents many have locomotion used to avoid predators

Nutritional modes in zooplankton Herbivores: feed primarily on phytoplankton Carnivores: feed primarily on other zooplankton (animals) Detrivores: feed primarily on dead organic matter (detritus) Omnivores: feed on mixed diet of plants and animals and detritus.

55.Steps Of Making Genomic Libraries? 5

steps for creating a genomic library from a large genome.

1. Extract and purify DNA.
2. Digest the DNA with a restriction enzyme. This creates fragments that are similar in size, each containing one or more genes.
3. Insert the fragments of DNA into vectors that were cut with the same restriction enzyme. Use the enzyme DNA ligase to seal the DNA fragments into the vector. This creates a large pool of recombinant molecules.
4. These recombinant molecules are taken up by a host bacterium by transformation, creating a DNA library. **(Internet)**

56.Write Down The Principal Of Gel Filtration Chromatography? 5

Gel filtration chromatography (sometimes referred to as molecular sieve chromatogra- phy) is a method that separates molecules according to their size and shape. The sepa- ration of the components in the sample mixture, with some exceptions, correlates with their molecular weights

57.Applications Of Plasmid Vectors? 5

Plasmids are the most-commonly used bacterial cloning **vectors**. These cloning **vectors** contain a site that allows DNA fragments to be inserted, for example a multiple cloning site or polylinker which has several commonly used restriction sites to which DNA fragments may be ligated

58.Advantage Of Yeast In Recombination?

Yeasts have been used for thousands of years in food and fermentation processes to produce alcoholic beverages and breads. In recent times, the industrial importance of yeasts has extended beyond its traditional use in fermentation into various healthcare sectors, such as in the production of therapeutic recombinant proteins. Yeast cells are particularly advantageous as hosts for biopharmaceutical production in that they are generally recognized as safe (GRAS) organisms.

59.What Are Gmos Example?

E.g., when a recombinant microbe is applied in bio augmentation it comes in competition with indigenous species which could outgrow it.

60.What Are Types Of Pollution?

Types of pollution There are different types of the pollution which are given below: Air pollution Water pollution Land pollution Environmental pollution Light pollution Noise pollution.

61.Enlist The Contaminants Released Into The Environment?

The chemicals or any thing which contaminate the environment is called pollutants. **(PPTS)**

contamination can result **from** a variety of intended, accidental, or naturally occurring activities and events such as manufacturing, mineral extraction, abandonment of mines, national defense activities, waste disposal, accidental spills, illegal dumping, leaking underground storage tanks, hurricanes, floods **(Internet)**

62.What Is Environmental Pollution?

Environmental Pollution: The presence of contaminants in the environment which produce the serious changes in the environment is called the environmental pollution. Pollutants: The chemicals or any thing which contaminate the environment is called pollutants.

63.Contaminants Released From The Environment?

Inorganic **pollutants**. Industrial, agricultural and domestic wastes contribute to **environmental** pollution, which cause adverse harm to human and animal health. From such sources, inorganic **pollutants** are **released**. **(Internet)**

64.Annealing Process?

annealing process, the metal is heated to a specific temperature where recrystallization can occur. ... The metal is held at that temperature for a fixed period, then cooled down to room temperature. The cooling **process** must be done very slowly to produce a refined microstructure, thus maximizing softness **(Internet)**

65.What Is Rnai Technology?

RNAi Technology. **RNA interference** (**RNAi**) is the process by which the translation of a protein is prevented by selective degradation of its encoded mRNA. In nature, this mechanism likely evolved for cells to eliminate unwanted foreign genes as a defense against viruses **(Internet)**

66.What Is Abiotic Stress?

The negative impact of the non-living factors on living organisms in a specific environment is called the abiotic stress.

67.What Is Annealing In PCR?

Polymerase chain reaction (PCR) The process is repeated by a constant cycling of denaturation of double stranded DNA at elevated temperature to approximately 95 ◦C, followed by cooling to approximately 55- 60 ◦C to allow annealing of the primers, followed by extension of the 3’ end of the primer by Polymerase usually at 72 ◦C.

68.What Is Aerobic Degradation And Anaerobic Degradations?

Aerobic Degradation: **AEROBIC BIODEGRADATION** Organic contaminants are rapidly It is the breakdown of organic contaminants by micro organism when oxygen is present **degraded** under **aerobic** conditions by **aerobic** bacteria called aerobes.**(Internet)**

Anaerobic Degradations: The aromatic hydrocarbons generally degrade readily under aerobic conditions, but anaerobic degradation by pure bacterial cultures has also been recognized **(PPTS)**

69.What Are The Two Schools Of Thought To Determine To Pursue The Biotechnologies?

Metaphysics attempts to find unity across the domains of experience and **thought**. At the metaphysical level, there are four* broad philosophical **schools of thought** that apply to education today. They are idealism, realism, pragmatism (sometimes called experientialism), and existentialism. Each will be explained shortly. **(Internte)**

70.Write A Note On Expression Vector?

These are similar to the vectors described above but in addition have the required regulatory regions located before and after the 'foreign' gene which direct the host cell to translate the product of transcription into a protein. It is sometimes a difficult, expensive or time consuming procedure to analyse for product from the 'foreign' gene and so, in addition to the selector genes described above, there are frequently reporter genes to indicate whether or not the signals are 'switched on' allowing the 'foreign' DNA to be expressed. In expression cloning, there are many circumstances which can be difficult to predict.

71.Difference Between Bioremediation & Environmental Biotechnology?

Bioremediation: **Bioremediation**. "Remediate" means to solve a problem, and "bio-remediate" means to use biological organisms to solve an **environmental** problem such as contaminated soil or groundwater. In a nonpolluted **environment**, bacteria, fungi, protists, and other microorganisms are constantly at work breaking down organic matter.

Environmental Biotechnology: **Environmental biotechnology** is the branch of **biotechnology** that addresses **environmental** problems, such as the removal of pollution, renewable energy generation or biomass production, by exploiting biological processes.

72.How To Generate Recombinant Virus Vaccine?

A number of genetically stable, safe live viral vaccines have been produced, e.g. measles and yellow fever. Live vaccines have a number of advantages but when attempting to expand the range of vaccines to include new targets it is often not possible to achieve a suitable live vaccine derivative of the pathogenic virus. Recombinant viral vaccines offer a path to achieving the benefits of live vaccines for such difficult cases. The principle is to embed one or more genes from the target pathogen into the genome of an established attenuated viral vaccine strain or other nonpathogenic viral genome Generate a Recombinant Virus Vaccine 198 Recombinant Viruses for Gene.

73.Define Biosenser ?

Biosensor

Environmental biosensor is the analytical device composed of biological sensing elements and a physically transducer which together relate to measure able signal

Types of biosensor based on transduction signal

- Electrochemical
- Optical
- Piezoelectric
- Thermal sensor

74.Examples of Endocrine disrupter?

Endocrine disruptor is:

An exogenous agent that interferes with synthesis, secretion, transport, binding, action, or elimination of natural hormones in the body which are responsible for the maintenance of homeostasis, reproduction, development and/or behavior

Examples of Endorcine Disruptors

- Diethylstilbosterol (DES).
- Polychlorinated biphenyls (PCBs).
- Polybrominated biphenyls (PBB).
- Chloroform.
- Dioxins and furans.
- Pesticides

75.Define endocrine disrupter?

Endocrine disruptor is:

An exogenous agent that interferes with synthesis, secretion, transport, binding, action, or elimination of natural hormones in the body which are responsible for the maintenance of homeostasis, reproduction, development and/or behavior.

Examples of Endorcine Disruptors

- Diethylstilbosterol (DES).
- Polychlorinated biphenyls (PCBs).
- Polybrominated biphenyls (PBB).
- Chloroform.
- Dioxins and furans.
- Pesticides

Mechanism of action

Endocrine disruptors act on the endocrine system of humans and animals by mimicking, blocking and/or interfering in some manner with the natural instructions of hormones to cells.

Binding of EDs to hormone receptors, with resultant alteration in the transcription of messenger RNA, followed by altered gene expression, appears to be one mechanism by which EDs can alter endocrine function.

Other mechanisms involve alteration in hormone synthesis, metabolism, and transport as well as effects mediated through changes in the hypothalamic– pituitary–gonadal axis.

76.Steps of making genimic labraries?

Steps of making Genomic Libraries

- Isolation of genomic DNA
- Generation of suitable size DNA fragments
- Cloning in suitable vector system (depending on size) •
  Transformation in suitable host.

Isolation of genomic DNA- Isolation of genomic DNA has following steps:

1. Lysis of cells with detergent containing lysis buffer.

2. Incubation of cells with digestion buffer containing protease-K, SDS to release genomic DNA from DNA-protein complex.

3. Isolation of genomic DNA by absolute alchol precipitation.

4. Purification of genomic DNA with phenol:chloroform mixture. Chloroform : phenol mixture has two phases, aqueous phase and organic phase. In this step, phenol denatures the remaining proteins and keep the protein in the organic phase.

5. Genomic DNA present in aqueous phase is again precipitated with absolute alchol.

6. Genomic DNA is analyzed on 0.8% agarose gel and a good preparation of genomic DNA give an intact band with no visible smear.

77.Define vector?

Cloning Vectors

A vehicle DNA is required to carry foreign DNA to generate a recombinant construct so that it should allow easy amplification of chimeric DNA in host. Vehicle DNA used for this purpose in genetic engineering is known as ‘vector’.

Criteria of a good vector :

• The vector DNA has a two main responsibility:
• Ability to carry foreign DNA,
• Ability to replicate in the host

78.xenobiotics,

XENOBIOTICS' are the foreign chemical substances found within an organism, which are not normally naturally produced by an organism or expected to be present within that organism.

• It can also cover substances which are present in much higher concentrations than are usual in any product.

• Most of the systems in the body are affected by these xenobiotic compounds including the immune, neurological, and the endocrine systems.

• Toxicity in these systems can lead to immune dysfunction, autoimmunity, asthma, cancer, learning difficulties, mood changes, neurological illnesses, changes in sexual desire and functioning, reproductive dysfunction, and upsets in blood glucose levels, etc.

79.what do u mean by mobility of DNA ,

The movement of genes within and between organisms.

• Plasmids may be transferred between bacteria by conjugation, of which there are several types, but all of which require direct cell to cell contact. Not only are genes transferred between bacteria on plasmids, but bacteriophages (bacterial viruses) are also vectors for intercellular transmission. Similarly, eukaryotic viruses are able to transfer genetic material between susceptible cells. In addition, bacterial cells may pick up DNA free in the environment under conditions where their cell wall has become 'leaky' to fragments of this macromolecule, a process called transformation.

80.define genetic engineering ,

Genetic Engineering

• Definition: The technology entailing all processes of altering the genetic material of a cell to make it capable of performing the desired functions, such as producing novel substances.

In other words:

• Genetic engineering is the deliberate, controlled manipulation of genes in an organism in order to upgrade that organism. Molecular tools and Technological breakthroughs

• Enzymes- nucleic acid cleavage, ligation

• Vector- molecular cloning

• Polymerase chain reaction

• DNA sequencing

• Electrophoretic separation

• Detection of genes:

DNA-Southern blotting; in situ hybridization;

FISH technique;

RNA- Northern blotting

Pr-Western blotting; inmmunohistochemistry

• Purification

• Transgenetic organisms

81.difference between exteremophiles and hyperthermophiles,

Extremophiles are organisms that have evolved to survive in environments once thought to be entirely uninhabitable. These environments are inhospitable, reaching extreme conditions of heat, acidity, pressure,

and cold that would be fatal to most other life forms. Because extremophiles live on extreme ends of the spectrum, they can indicate the range of conditions under which life is possible.

Read more on Brainly.in -

A hyperthermophile is an organism that thrives in extremely hot environments—from 60 °C (140 °F) upwards. An optimal temperature for the existence of hyperthermophiles is above 80 °C (176 °F).[1]Hyperthermophiles are often within the domain Archaea, although some bacteria are able to tolerate temperatures of around 100 °C (212 °F), as well. Some bacteria can live at temperatures higher than 100 °C at large depths in sea where water does not boil because of high pressure. Many hyperthermophiles are also able to withstand other environmental extremes such as high acidity or high radiation levels. Hyperthermophiles are a subset of extremophiles

contaminants releases in the environment.

Contaminants released into the environment

- Solvents
- Pesticides
- Herbicides
- Fungicides
- Insecticides
- Petrochemicals
- Explosives
- Heavy metals

Environment biotechnology deals with

- Decontamination of environmental components
- Production of chemicals biosensors
- Pollution prevention
- Waste minimization

82.what do you mean by mobilioty of DNA? annealing steps in pcr?

Annealing: This is the annealing step where at lower temperature (usually 50-650C) primers are allowed to bind to template DNA, annealing time is 15-30 secs and it depends on the length and bases of the primers. Generally annealing temperature is about 3-5°C below the melting temperature (Tm) of the pair of the primers is used.

83.RNAi technology? 3 marks

RNAi technology

• In addition to the traditional strategies for vector construction and genetic modification strategies described, RNAi (interference) technology is now becoming a new way to improve the contents and fight the diseases of crop plants.

• Furthermore, plant virus vectors for transfer and expression of transgenes in crop plants are coming into use.

84.pU38 plasmid? 3 marks

The pUC8 plasmid was designed by scientists and contains the lac z gene. To produce the plasmid, the pBR322 plasmid was cut in half with EcoR I and the section containing the ampicillin resistance gene was combined with a DNA fragment containing the lac z gene.

• As a result, the plasmid provides a transformed cell with both ampicillin resistance and the ability to utilize lactose as a food source, since the lac z gene produces B galactosidase (degrades lactose)

85.What are cosmid vectors (3)

Cosmids vector

- Combine parts of the lambda chromosome with parts of plasmids.
- An origin of replication (ori).

- A cos site(a sequence yield cohesive end) .
- An ampicillin resistance gene (amp),
- Restriction sites for cloning
- Cosmids can carry up to 50 kb of inserted DNA.

86.Applications of plasmid vectors (5)

A plasmid is a small DNA molecule within a cell that is physically separated from a chromosomal DNA and can replicate independently. They are most commonly found as small circular, double-stranded DNA molecules in bacteria; however, plasmids are sometimes present in archaea and eukaryotic organisms. In nature, plasmids often carry genes that may benefit the survival of the organism, for example antibiotic resistance. While the chromosomes are big and contain all the essential genetic information for living under normal conditions, plasmids usually are very small and contain only additional genes that may be useful to the organism under certain situations or particular conditions. Artificial plasmids are widely used as vectors in molecular cloning, serving to drive the replication of recombinant DNA sequences within host organisms. In the laboratory, plasmids may be introduced into a cell via transformation.

Plasmids are considered replicons, units of DNA capable of replicating autonomously within a suitable host. However, plasmids, like viruses, are not generally classified as life.[1] Plasmids are transmitted from one bacterium to another (even of another species) mostly through conjugation.[2] This host-to-host transfer of genetic material is one mechanism of horizontal gene transfer, and plasmids are considered part of the mobilome. Unlike viruses (which encase their genetic material in a protective protein coat called a capsid), plasmids are "naked" DNA and do not encode genes necessary to encase the genetic material for transfer to a new host.

However, some classes of plasmids encode the conjugative "sex" pilus necessary for their own transfer. The size of the plasmid varies from 1 to over 200 kbp,[3] and the number of identical plasmids in a single cell can range anywhere from one to thousands under some circumstances.

The relationship between microbes and plasmid DNA is neither parasitic nor mutualistic, because each implies the presence of an independent species living in a detrimental or commensal state with the host organism. Rather, plasmids provide a mechanism for horizontal gene transfer within a population of microbes and typically provide a selective advantage under a given environmental state. Plasmids may carry genes that provide resistance to naturally occurring antibiotics in a competitive environmental niche, or the proteins produced may act as toxins under similar circumstances, or allow the organism to utilize particular organic compounds that would be advantageous when nutrients are scarce